

The Sharp Lipschitz Exponent for Ball-PSWF Heat Kernels

When $\gamma \geq 0$

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Status. Substantial partial result, likely valid, pending human review. The exponent in the source question is proved to be the maximal value $\alpha = 1$ for every $\gamma \geq 0$. The range $-\frac{1}{2} < \gamma < 0$ remains open here.

1 Source question and result

Bonami, Kerkycharian and Petrushev consider on the unit ball $\mathbb{B}^d$, $d \geq 2$, the measure

$$d\mu_\gamma(x) = (1 - |x|^2)^{\gamma-1/2} dx, \quad \gamma > -\frac{1}{2},$$

and the positive ball-PSWF operator

$$L_c = -\Delta + \sum_{i,j=1}^d x_i x_j \partial_i \partial_j + (d + 2\gamma)x \cdot \nabla + c^2|x|^2.$$

Its intrinsic distance is

$$\rho(x, y) = \arccos\left(x \cdot y + \sqrt{1 - |x|^2} \sqrt{1 - |y|^2}\right).$$

Writing $p_t^c(x, y)$ for the heat kernel of e^{-tL_c} and $V(x, r) = \mu_\gamma(B_\rho(x, r))$, the source proves that some $0 < \alpha \leq 1$ satisfies its Gaussian Hölder estimate (8.23), and asks for the dependence of α on γ [1, p. 40]:

Theorem 1 (Nonnegative-parameter answer). *Let $d \geq 2$, $\gamma \geq 0$, and $c \geq 0$. There are constants $C, c_0 > 0$ such that, whenever $\rho(x, x') \leq \sqrt{t}$,*

$$|p_t^c(x, y) - p_t^c(x', y)| \leq C \frac{\rho(x, x')}{\sqrt{t}} \frac{\exp(-\rho(x, y)^2/(c_0 t))}{[V(x, \sqrt{t})V(y, \sqrt{t})]^{1/2}}. \quad (1)$$

Thus the exponent in the source question can be chosen to be

$$\boxed{\alpha = 1}$$

throughout the range $\gamma \geq 0$.

Since no Hölder exponent larger than 1 is under consideration, this is the sharp value of the exponent on this parameter range.

Therefore, $\bar{\mathcal{E}}$ is a Dirichlet form. Now, the theory from Section 2 can be applied. In particular, estimates (8.10) follow from Theorem 2.1 and the Gaussian bounds for $\mathcal{K}_{\exp(-tL_0)}(x, y)$. $\square$

8.3. Hölder continuity of the PSWF heat kernel $p_t(x, y)$ on $\mathbb{B}^d$. As is shown in [35] the quadratic form $\mathcal{E}$ from (8.21) generates a regular strictly local Dirichlet space and the respective Poincaré inequality is satisfied. This together with the doubling property of the measure gives another proof of the Gaussian bounds for $\mathcal{K}_{\exp(-tL_0)}(x, y)$. In turn, this yields the Hölder continuity of the kernel $\mathcal{K}_{\exp(-tL_0)}(x, y)$. Furthermore, the Hölder continuity of $\mathcal{K}_{\exp(-tL_0)}(x, y)$ and (8.10), in light of Theorem 2.2, imply the Hölder continuous of the PSWF heat kernel $p_t(x, y)$ on $\mathbb{B}^d$: There exists a constant $0 < \alpha \leq 1$ such that

$$(8.23) \quad |p_t(x, y) - p_t(x', y)| \leq c_1 \left(\frac{\rho(x, x')}{\sqrt{t}} \right)^\alpha \frac{\exp\left(-\frac{\rho(x, y)^2}{c_2 t}\right)}{[V(x, \sqrt{t})V(y, \sqrt{t})]^{1/2}},$$

whenever $\rho(x, x') \leq \sqrt{t}$. Here α is the same as for $\mathcal{K}_{\exp(-tL_0)}(x, y)$. It is an open question to find the dependence of α on γ .

8.4. Besov and Triebel-lizorkin spaces associated to the PSWFs on $\mathbb{B}^d$. The theory of Besov and Triebel-lizorkin spaces on $\mathbb{B}^d$ associated to the operator L_c in the setting described above is now easy to develop due to the fact that the Gaussian bounds of the heat kernel (8.10) and its Hölder continuity (8.23) are already established. These allow to develop the relevant smooth functional calculus and then the associated Besov and Triebel-lizorkin spaces with a complete set of

Figure 1: Equation (8.23) and the open question in [1, p. 40].

2 Proof intuition

The source's distance is not ad hoc: the map

$$x \mapsto (x, s(x)), \quad s(x) = \sqrt{1 - |x|^2},$$

identifies the ball with the upper hemisphere. Under this map, the ball measure becomes spherical measure weighted by $s^{2\gamma}$, and $-L_0$ is the corresponding weighted Laplacian.

For $\gamma \geq 0$ this weighted sphere has nonnegative Bakry-Émery curvature. The resulting reverse Poincaré inequality bounds the intrinsic gradient of the heat kernel by $t^{-1/2}$ times a Gaussian kernel profile. Integrating the gradient along the short spherical geodesic from x to x' gives the Lipschitz factor $\rho(x, x')/\sqrt{t}$. The source paper's bounded-potential transfer theorem then carries exactly the same exponent from L_0 to $L_c = L_0 + c^2|x|^2$.

3 The weighted hemisphere

Let

$$H_+^d = \{(x, s) \in \mathbb{S}^d : s > 0\}, \quad s = s(x) = \sqrt{1 - |x|^2}.$$

The pullback of spherical distance is precisely ρ . Spherical volume in the x chart is $d\text{vol}_{\mathbb{S}} = s^{-1}dx$, and hence

$$s^{2\gamma} d\text{vol}_{\mathbb{S}} = (1 - |x|^2)^{\gamma-1/2} dx = d\mu_\gamma(x). \quad (2)$$

The weighted spherical diffusion is

$$\mathcal{L}_\gamma = \Delta_{\mathbb{S}^d} + 2\gamma \nabla \log s \cdot \nabla.$$

In the x chart,

$$\mathcal{L}_\gamma = \sum_{i,j} (\delta_{ij} - x_i x_j) \partial_i \partial_j - (d + 2\gamma) x \cdot \nabla = -L_0.$$

Thus the unperturbed heat kernel from the source is the heat kernel of this weighted hemisphere diffusion (with its natural weighted-Neumann closure).

Put $e^{-W} = s^{2\gamma}$, so $W = -2\gamma \log s$. On the unit sphere,

$$\text{Ric} = (d-1)g, \quad \text{Hess}(\log s) = -g - \frac{ds \otimes ds}{s^2}.$$

Therefore its Bakry–Émery tensor is

$$\text{Ric}_W = \text{Ric} + \text{Hess } W = (d-1+2\gamma)g + 2\gamma \frac{ds \otimes ds}{s^2} \geq 0 \quad (3)$$

for every $\gamma \geq 0$. Equivalently, the diffusion satisfies $\text{CD}(0, \infty)$. Formula (3) can also be checked directly on the polynomial core by the Bochner identity. The associated gradient inequality and its reverse-Poincaré consequence pass to the closed Markov semigroup; see [2, Chapters 3–4]. At $\gamma = 0$ this is simply the ordinary spherical semigroup restricted to functions even across the equator.

4 From curvature and Gaussian bounds to Lipschitz bounds

We isolate the short argument that retains the Gaussian localization. Let $P_t = e^{-tL_0}$, let q_t be its heat kernel, and write $\Gamma(u) = |\nabla u|_g^2$. The $\text{CD}(0, \infty)$ reverse Poincaré inequality is

$$2a\Gamma(P_af) \leq P_a(f^2) - (P_af)^2 \leq P_a(f^2). \quad (4)$$

Fix $t > 0$ and y , put $a = t/2$, and apply (4) to $f(z) = q_{t/2}(z, y)$. Positivity and the semigroup identity give

$$\begin{aligned} \Gamma_x(q_t(x, y)) &\leq \frac{1}{t} P_{t/2}(q_{t/2}(\cdot, y)^2)(x) \\ &\leq \frac{1}{t} \|q_{t/2}(\cdot, y)\|_\infty q_t(x, y). \end{aligned} \quad (5)$$

The Gaussian upper estimate already proved in the source, together with doubling, has the standard asymmetric consequences

$$\|q_{t/2}(\cdot, y)\|_\infty \leq \frac{C}{V(y, \sqrt{t})}, \quad q_t(x, y) \leq \frac{C}{V(y, \sqrt{t})} e^{-\rho(x, y)^2/(Ct)}.$$

Taking square roots in (5) yields

$$|\nabla_x q_t(x, y)|_g \leq \frac{C}{\sqrt{t} V(y, \sqrt{t})} e^{-\rho(x, y)^2/(Ct)}. \quad (6)$$

Let $\delta = \rho(x, x') \leq \sqrt{t}$ and integrate (6) along the minimizing spherical geodesic joining x to x' . Along this segment, the triangle inequality preserves the Gaussian factor after enlarging its denominator constant. Hence

$$|q_t(x, y) - q_t(x', y)| \leq C \frac{\delta}{\sqrt{t} V(y, \sqrt{t})} e^{-\rho(x, y)^2/(Ct)}.$$

Doubling gives, for some $Q > 0$,

$$V(x, r) \leq C \left(1 + \frac{\rho(x, y)}{r}\right)^Q V(y, r).$$

The polynomial factor is absorbed by the Gaussian, converting the last display to

$$|q_t(x, y) - q_t(x', y)| \leq C \frac{\rho(x, x')}{\sqrt{t}} \frac{e^{-\rho(x, y)^2/(Ct)}}{[V(x, \sqrt{t})V(y, \sqrt{t})]^{1/2}}. \quad (7)$$

Finally, $L_c = L_0 + c^2|x|^2$ is a bounded nonnegative potential perturbation. The source's Hölder-transfer theorem [1, Theorem 2.2] preserves the exponent in (7); its application in Section 8.3 is precisely what produces equation (8.23). It therefore gives (1) with $\alpha = 1$, proving Theorem 1. $\square$

5 Verification, limitations, and novelty boundary

The proof is analytic and has no computational dependency. The generator calculation agrees term by term with the source's L_0 . The key sign is in (3): both additional terms are nonnegative exactly when $\gamma \geq 0$. The kernel substitution in (5) then uses only positivity, the semigroup law, the source's Gaussian bound, and doubling.

For $-\frac{1}{2} < \gamma < 0$, the normal curvature term $2\gamma ds \otimes ds/s^2$ tends to $-\infty$ at the equator. Thus this proof supplies no uniform reverse Poincaré estimate. The regular branch of the one-dimensional Bessel boundary model suggests that $\alpha = 1$ may still be true, but a global scale-invariant gradient estimate was not obtained. Eight focused attempts, including parameter-shift and spectral-kernel routes, are recorded in the attempt log.

Bounded searches on 17 August 2026 covered the run indexes, exact source metadata and question wording, weighted-ball gradient and Lipschitz phrases, and the published source's citation list. Its sole OpenAlex-listed citing paper is unrelated to this exponent. No explicit later determination was found. Novelty confidence is moderate because the weighted curvature method is standard once the hemisphere identification is made.

Human review is recommended, focusing on the Markov closure at the equator for $0 < \gamma < 1/2$, the pointwise use of reverse Poincaré, and the final application of the source's perturbation theorem.

References

- [1] A. Bonami, G. Kerkycharian and P. Petrushev, *Gaussian bounds for the heat kernel associated to prolate spheroidal wave functions with applications*, Constr. Approx. 56 (2022), 517–567; arXiv:2108.12917.
- [2] D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren der mathematischen Wissenschaften 348, Springer, 2014.